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Governing Credit in the Digital Age: Public Perceptions and Engagement in China's Credit Systems

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ABSTRACT

There is a global trend toward embedding personal credit systems and their scoring mechanisms within broader governance infrastructures. A prominent and controversial example is China's Social Credit System (SCS), which plays a central role in the country's data-driven financial and social governance. This study examines how Chinese citizens conceptualize "credit" and "social credit" and to what extent they engage with various personal credit systems, drawing on a large-scale survey ($N=5538$). Our findings reveal a strong emphasis on the social attribute of both "credit" and "social credit," with respondents perceiving a close relationship between personal credit and reputation. This perception highlights the growing convergence of financial and reputational metrics. We observe significant variations across systems: participants are most engaged with commercial credit scoring systems, while they exhibit the least engagement with administrative credit rating systems. Notably, income emerges as a strong predictor of engagement, suggesting that wealthier individuals tend to engage more actively with personal credit rating systems and may derive more perceived benefits. This disparity raises the possibility of deepening financial and economic inequalities. By contextualizing China's experience within broader discussions of algorithmic regulation and data-driven financial systems, this study contributes to the ongoing discourse on digital inequality, algorithmic power, and financial stratification in non-Western contexts.

1 | Introduction

In the digital era, the datafication of social and economic behaviors has led to a global trend: the increasing use of data-driven scores for regulatory and governance purposes (Dencik et al. 2018). These scores, derived from diverse data sources, such as online activities, financial and legal records, are now employed to assess and influence individual and organizational behaviors across various sectors. China's Social Credit System (SCS) exemplifies this broader trend, leveraging the concept of personal credit as a tool for broader social regulation. Unlike the well-established credit systems in the West, which typically focus on financial reliability for lending purposes (Abdou and Pointon 2011), China's SCS, as an overarching governance framework, coordinates diverse financial, social, and administrative credit mechanisms to shape behaviors beyond financial transactions (Chen and Grossklags 2022; Creemers 2018).

China's credit system developed relatively late, with enterprise credit reporting being initiated in the late 1980s and personal credit reporting emerging in 1999 (Jentzsch 2008). The former assesses the financial reliability of businesses, while the latter evaluates individual creditworthiness. Given the slow evolution of China's personal credit system, which could be attributed to a weak credit culture (Wei 1999) and a substantial unbanked population (of about 287 million individuals) (Baltrusaitis 2021), the SCS was introduced to accelerate its development. Nowadays, China has three main personal credit systems: financial credit reporting, commercial credit, and administrative credit (Wu 2020), all linked to or facilitated by the SCS (see Section 2). Thus, the SCS serves as an overarching framework for personal credit in China.

The SCS framework builds on state-led oversight and extensive use of digital infrastructures. Big data analytics and emerging technologies are employed to systematically assess, educate, and influence the social and economic activities of individuals and organizations (Creemers 2018; Liang et al. 2018). In the words of the Chinese government, the SCS aims to "allow the trustworthy to roam everywhere under heaven while making it hard for the discredited to take a single step" (State Council 2014).

The SCS has become increasingly integral to China's socioeconomic landscape, pursuing three main goals. First, it facilitates algorithmic regulation in social governance. As governments worldwide have begun to employ algorithmic regulation in the public sector (Yeung 2018), the SCS stands out as a large-scale and controversial case (Cristianini and Scantamburlo 2020; Mac Síthigh and Siems 2019; State Council 2014), with media and academia widely criticizing its surveillance nature, lack of transparency, and privacy infringements (Chen, Bogner, et al. 2023; Chen, Engelmann, and Grossklags 2023; Hoffman 2018; Lee 2019; Liang et al. 2018; Suter 2020). Second, the SCS purportedly aims to foster a culture of trust and trustworthiness in Chinese society (State Council 2014). Scandals in both financial (e.g., Evergrande accounting fraud) and non-financial sectors (e.g., 2008 milk scandal) exposed deep trust deficits. Further, a series of Ipsos surveys identified moral decline as a top concern in China,[1] highlighting the need for stronger social trust. Third, the SCS strives to improve financial inclusion and economic growth. Access to credit is crucial for financial well-being and economic development (Claessens 2006; Demirgüç-Kunt and Levine 2008). By assessing creditworthiness beyond financial records—incorporating administrative and moral behavior (Engelmann et al. 2021)—the SCS intends to provide unbanked people access to credit and investment opportunities.

However, as an algorithmic behavior evaluation system, the SCS also raises concerns about algorithmic control, data-driven social sorting, and the opacity of automated decision-making (Pasquale 2015; Verma 2019). The increasing reliance on AI-powered credit scoring and predictive analytics has intensified debates on transparency, fairness, and potential discrimination in algorithmic systems (Loefflad et al. 2024). Moreover, scholars have highlighted the distributive implications of these systems (Loefflad and Grossklags 2024), warning that they may reinforce or generate new forms of digital inequality (Büchi et al. 2016; van Deursen and van Dijk 2014).

Despite these important theoretical debates, empirical research on the nuanced public perceptions of these systems, people's engagement with them, and the systems' socio-economic impacts remains significantly underdeveloped, leaving a critical gap in our understanding of how these systems operate in practice. This gap is apparent within the context of the SCS. In this study, *engagement* is conceptualized as the extent of individuals' knowledge, information-seeking behaviors, and use of credit scores for benefits across different personal credit rating systems. While various personal credit rating systems are actively integrated in society through the SCS, it remains largely unclear how Chinese citizens interpret these fundamental concepts that inform the system. In particular, given the complex landscape, short history, and extensive use of digital technologies in China's personal credit systems, the coexistence of multiple systems may suggest conceptual distinctions between different types of credit. In addition, there are only a few empirical studies which primarily focus on public approval of the system, showing relatively high support (ranging from 41% to 80%) among Chinese citizens (Chen and Grossklags 2022; Kostka 2019; Li and Kostka 2022; Liu 2022; Ohlberg 2020), especially when contrasted with critical portrayals in Western media (Mosher 2019; Palin 2018). But there is a lack of systematic investigation of how people engage with different types of personal credit rating systems in daily life. Moreover, differences in access to digital technologies, skills in their use, information-seeking behavior, social support, and usage patterns may reflect or even actively reproduce existing social inequalities (Hargittai 2001; Warschauer 2004). Understanding how such engagement varies across social and demographic groups is, therefore, essential for assessing the broader societal implications of the SCS.

This study addresses these gaps using survey data from a representative Chinese sample ($N = 5538$) and employs a mixed-methods approach. Specifically, we address two questions. First, we explore how Chinese citizens conceptualize "credit" and "social credit." Then, we develop hypotheses relating to individuals' engagement across different personal credit rating systems, specifically regarding their knowledge, usage, and information-seeking behaviors.

Drawing on governmentality (Foucault 1991), policy feedback theory (Mettler and Soss 2004), social sorting (Lyon 2003), and digital inequality (Hargittai 2001), this study investigates how the SCS framework shapes public perceptions and engagement. Our work advances existing scholarship by shifting the analytical focus from institutional design to everyday engagement, highlighting how individuals interpret, internalize, or resist the governing logic embedded in algorithmic infrastructures. Through this perspective, personal credit systems are examined not merely as instruments of surveillance and regulation, but as relational arenas in which citizens actively negotiate inclusion, exclusion, and trust. In doing so, the study illuminates the distributive and behavioral implications of data-driven governance, contributing to broader debates on algorithmic regulation and digital inequality. Our work underscores that the socio-economic impacts of such systems are not determined solely by technological design or policy intent, but also by how people engage with, make sense of, and act within these emerging credit-based governance regimes.

2 | Personal Credit Systems in China: An Overview

As previously mentioned, China's personal credit systems are categorized into three main types. Both the underlying data and models used by the three systems to assess creditworthiness are distinct, each serving different purposes.

2.1 | The Financial Credit Reporting System

The financial credit reporting system in China is state-led, primarily assessing individuals' creditworthiness for loan and credit applications. The People's Bank of China (PBoC) initiated the construction of a national financial credit reporting system in 2004, which was then launched 2 years later (Chen and Grossklags 2020). Nowadays, it is the world's largest personal credit system, covering 1.15 billion individuals.[2] Credit reports include basic information (e.g., name, address, contact details, marital status, occupation), transaction history (e.g., records related to credit cards, loans and other financial activities), public records (social security and provident fund information, administrative information, records in the courts, etc.), and credit inquiries in the past 2 years. A "numerical interpretation (数字解读)" score (0–1000) indicates default risk, where higher scores signify lower risk. A "relative ranking (相对位置)" percentile compares an individual to the broader population. Similar to credit reports from Experian (in the U.S.) and Schufa (in Germany), PBoC reports are commonly utilized for financial services, and also during job searches, apartment rentals, and even for international travel. China's financial credit reporting system operates under a strict licensing framework. Three private entities, Baihang Credit, Pudao Credit, and Qiantang Credit, licensed and supervised by the PBoC,[3] were established in 2018, 2020 and 2024, respectively. Jointly owned by state and private investors, these agencies serve as supplements to the PBoC's Credit Reference Center but have a much smaller coverage scope: as of the end of 2023, Baihang Credit and Pudao Credit held records on roughly 669 and 620 million individuals, respectively.[4]

Taken together, these institutional features also have important social implications. By evaluating socio-economic characteristics jointly with financial information, the financial credit reporting system understands credit as a quantified assessment of one's socio-economic position in society. As a reputational signal, it informs about individuals' creditworthiness and economic trustworthiness. In this sense, financial credit functions as an instrument of economic stratification, which makes distinctions in socio-economic status visible, and thus shapes access to financial opportunities. It reflects a structural power in economic relations through a socio-economically informed notion of credit.

2.2 | The Commercial Credit System

China's personal credit system began marketization in 2015 when eight commercial companies, including Zhima Credit, Tencent Credit, Qianhai Credit, Pengyuan Credit, China Chengxin Credit, IntelliCredit, Sinoway Credit, and Koala Credit, were granted permission to operate personal credit reporting services. However, after a two-year trial, none received official licenses. Some, like China Chengxin Credit, exited the sector to focus on enterprise credit, while others integrated credit data into their business models (Wu 2020), fostering the growth of commercial credit systems. Zhima Credit (or Sesame Credit) affiliated with Alibaba, exemplifies this shift by integrating individuals' Zhima Score into various online services and businesses within Alibaba's ecosystem and beyond. It utilizes cloud computing and machine learning to assess commercial creditworthiness based on four key factors: compliance records (e.g., court enforcement records, contract fulfillment), behavior accumulation (e.g., shopping, mobile payments, utility payments, credit card repayments, and participation in public welfare), identity verification (e.g., education, profession, identity card, driving license, and passport), and asset proof (e.g., real estate, vehicle assets, social security, housing provident fund, personal income tax, and financial products).[5] Zhima Credit assigns scores from 350 to 950, with higher scores reflecting stronger creditworthiness. These scores make benefits available during shopping, car rental, hotel booking, apartment leasing, and even for the use of public services like healthcare and visa applications. With a user base that likely exceeds 1 billion individuals,[6] Zhima Credit is one of China's most widely used personal credit services.

The commercial system uses credit as a basis for transactions and economic exchanges, where credit scores are used as a reputational currency that complements more traditional notions of financial credit. As a basis for accessing benefits and services, the commercial system highlights credit as a tool for material exchange. Prior research, however, has claimed that the commercial SCS is only of limited functional use (Kostka and Antoine 2019), which implies that the system does not encompass a broader regulation or governance strategy.

2.3 | The Administrative Credit System

The administrative credit system refers to the government-run SCS, aiming to support administrative oversight and public service (Wu 2020). Often described as a reputation-based system (Dai 2018; Mac Síthigh and Siems 2019), it relies on blacklists and redlists to record "blameworthy" and "praiseworthy" behavior of individuals and companies, encompassing both legal compliance and ethical conduct. The entries usually include name, partially anonymized ID number, and reasons for inclusion. They are publicly accessible for shaming and praising purposes. The SCS is implemented at multiple levels. Nationally, it is overseen by the PBoC and the National Development Reform Commission (NDRC), with the State Council issuing high-level policies and regulations (e.g., (State Administration for Market Regulation 2022; State Council 2014, 2022)). In response, provinces and cities develop and enforce localized policies, and various government agencies use distinct approaches to blacklist, redlist, rate, or score individuals and companies (Chen and Grossklags 2025). There is currently no unified national credit score for individuals. Instead, over 50 local governments have created independent personal credit scoring systems, assessing residents' social, economic and moral behavior (Li and Kostka 2022). For instance, in many cities, donating to charity and volunteering can improve an individual's credit score, while unpaid utility bills or administrative punishments may lower it. Although residents can choose whether to activate their personal credit score via local apps or use associated services, all citizens in the targeted population are assessed by these systems. That is, participation in the digital interface is voluntary, but the underlying evaluation and credit assessments are mandatory and enforced by administrative agencies. Higher scores can provide benefits such as discounted parking and gym services or no-deposit library access. However, scoring criteria, calculation methods, and rewards vary by location, and many cities have yet to introduce a personal credit scoring system for residents. The administrative credit system thus extends beyond financial or commercial metrics, combining administrative compliance and moral behavior to regulate access to public services and benefits.

The administrative system is rooted in social governance and moral regulation, and stresses legal compliance and a state-defined notion of ethical conduct in society. It uses credit to evaluate individuals' moral and social contributions to society, where credit is supposed to give indications about one's moral value and social trustworthiness. In the administrative system, social credit expresses the government's symbolic power over citizens (Kostka and Antoine 2019), reinforcing state norms by linking behavioral expectations to access to public services and reputational standing.

2.4 | Gaps in Existing Research

In summary, individuals based in China are exposed to a variety of credit systems that embed different notions of credit and distinct underlying themes. Yet, how individuals themselves understand credit and social credit remains unexplored. Moreover, assessing their levels of engagement with these systems provides insight into whether individuals adopt and accept the underlying themes associated with the broader concept of credit. Understanding these questions helps clarify how credit is interpreted and functions from a citizen perspective, and whether it is accepted and engaged with as a fundamental regulatory and governance tool. Lastly, examining which socio-demographic groups engage with the different systems offers insights into the broader societal implications of these credit practices, including potential inequalities stemming from a differential adoption of credit infrastructures.

Existing studies offer fragmented insights into public engagement with these systems. For instance, Li and Kostka (2022) examine engagement in the local SCS but overlook usage patterns and other types of personal credit systems. Kostka and Antoine (2019) analyze behavioral adaptations under the SCS but neglect knowledge of the system as a prerequisite for meaningful engagement. These works leave critical questions unresolved, including which demographics understand, use, and seek information through these systems, and whether behavioral shifts reflect informed awareness or passive conformity.

Our study addresses these gaps by investigating the public understanding of the concepts "credit" and "social credit." We further systematically assess engagement across all three system types. We define engagement as comprising three interrelated dimensions: (1) *knowledge* of the system, (2) *information-seeking behaviors*, and (3) *usage* of scores for tangible benefits. While knowledge is often categorized as a purely cognitive attribute in classical public opinion research (e.g., (Carpini and Scott 1996)), this distinction is less applicable in the context of algorithmic governance (Ananny and Crawford 2018). Knowledge constitutes an epistemic dimension of engagement with power infrastructures, providing the informed awareness required for meaningful interaction with complex algorithmic systems. Drawing on policy feedback theory, we thus conceptualize knowledge as a constitutive dimension of engagement that shapes how citizens relate to and form orientations toward data-driven governance (Mettler and Soss 2004; Pierson 1993). Information-seeking behaviors represent a behavioral effort to render the opaque logic of algorithmic governance legible. By actively seeking information, individuals move beyond being passive subjects of data collection to become active auditors of their digital standing, thereby exercising a form of agency within the credit infrastructure (Ananny and Crawford 2018). Usage signifies the practical application of credit scores to secure tangible personal benefits, representing the process of turning one's credit into real-world opportunities. Accordingly, knowledge, information-seeking, and usage form a joint process in which knowing the system constitutes a foundational act of participation that enables intentional interaction and structures citizens' relationship with the regulatory regime (Büchi et al. 2016).

China's personal credit landscape provides a distinctive non-Western context for examining how data infrastructures mediate governance and citizenship. While Western credit regimes typically focus on market-based financial risk assessment, China's integrated approach merges financial, commercial, and administrative rationalities. This convergence expands the scope of algorithmic governance from economic behavior to moral and civic domains, blurring the boundaries between regulation, reputation, and engagement. Recognizing these contextual specificities is essential for theorizing how citizens make sense of credit, negotiate digital inclusion, and respond to new forms of socio-technical regulation.

The following section derives hypotheses regarding: relative engagement across different types of systems, engagement variations between cities with/without a local scoring system, and socio-demographic disparities in engagement.

3 | Hypotheses

Drawing on policy feedback theory (Mettler and SoRelle 2018; Mettler and Soss 2004; Pierson 1993), empirical research on public attitudes toward the SCS (Chen and Grossklags 2022; Kostka 2019; Li and Kostka 2022; Liu 2022; Ohlberg 2020), and studies on credit knowledge and behavior (Bernerth 2012; Borden et al. 2008; Levinger et al. 2011; Robb and Woodyard 2011), we develop hypotheses to examine how institutional, local, and social mechanisms shape individuals' engagement with personal credit systems in China. Policy feedback theory suggests that policy designs influence citizens' knowledge, attitudes, and behaviors by structuring their interactions with governing institutions. Within this framework, financial, commercial, and administrative credit systems embody distinct institutional logics and feedback mechanisms that shape how citizens perceive and utilize them. Moreover, comparing cities with and without a local social credit scoring system highlights the localized nature of algorithmic governance, revealing how exposure to government-led digital infrastructures affects individuals' engagement with these systems. Finally, socio-demographic differences capture enduring digital inequalities and unequal capacities for civic and economic participation.

Taken together, these three dimensions—institutional, local, and social—constitute a theoretically grounded framework for analyzing variation in knowledge, usage, and information-seeking behaviors across China's personal credit systems. Accordingly, we develop hypotheses to examine differences in individuals' engagement with personal credit systems across: (1) credit system types, (2) cities with and without social credit scoring systems, and (3) socio-demographic groups. To maintain analytical coherence, the hypotheses are organized into three clusters corresponding to these dimensions.

The coexistence of these three systems illustrates the hybrid and multi-layered nature of data-driven governance in China. Each system embodies a distinct institutional logic: the financial system centers on regulatory compliance and market stability; the commercial system translates digital behaviors into measures of creditworthiness and trust; and the administrative system operationalizes moral and civic evaluations as tools of governance. Understanding how citizens navigate, interpret, and engage with these overlapping systems is crucial, as each offers different incentives, interfaces, and degrees of transparency (Chen and Grossklags 2020; Engelmann et al. 2019). These structural differences are likely to shape individual engagement across different systems (Loefflad et al. 2025). In the following, we distinguish between two groups of cities: C1 representing cities that have implemented a local credit scoring system, and C2 representing cities that have not.

3.1 | System-Based Engagement Patterns

This section posits hypotheses concerning differences in citizen engagement across the three distinct personal credit systems examined in this study: commercial credit systems (e.g., Zhima), financial credit systems (PBoC), and administrative credit systems (the national SCS and local credit scoring systems). Empirical findings indicate variations in citizens' engagement with different systems in China: while the commercial credit system in China has the highest adoption rate (80.4%), with Zhima Credit being the most popular one (58.2%) (Kostka 2019), public awareness of and enrollment in administrative credit systems is very low, both at the national and the local levels (Chen and Grossklags 2022; Drinhausen and Brussee 2021; Kostka 2019; Li and Kostka 2022; Tsai et al. 2021). The PBoC credit reports are usually overlooked in existing literature. Despite covering data on 1.15 billion individuals—comparable in scale to Zhima Credit—they receive significantly less user engagement, with an average of 10.84 million daily visits compared to Zhima Credit's 650 million monthly active users.[7] However, the above mentioned existing research typically examines only isolated aspects of engagement, such as knowledge of the SCS or participation in a local scoring system, rather than providing a systematic, comparative account of engagement across all three systems and across multiple behavioral dimensions. Against this background, we hypothesize systematic differences in engagement across the three dimensions:

H1. *System dominance. Citizens exhibit the highest overall engagement (knowledge, information seeking, and use of score for benefits) with commercial credit systems and the lowest with administrative credit systems, even in cities with a local administrative scoring system* (C1).

3.2 | Local Administrative System Presence and Engagement Patterns (C1 vs. C2)

This section develops a hypothesis regarding whether engagement with personal credit systems is affected by the presence of local administrative scoring systems. To this end, we compare engagement patterns of citizens from cities with a local credit scoring system (C1) with those of citizens from cities without such a system (C2). The primary differentiator between these two groups is the implementation of a local credit scoring system, which functions as an important local governance infrastructure. The civic voluntarism model posits that public policies influence mass political behavior in various ways (Verba et al. 1995). Research on policy feedback shows that the design, implementation, and distribution of policy benefits can influence how people think and act politically (Mettler and Soss 2004; Pierson 1993; Skocpol 1995). These influences generally occur through two mechanisms: resource effects and interpretive effects. Resource effects arise when policies provide material benefits, such as financial support, services, and goods that encourage citizens to participate more actively in civic and political life (Campbell 2002). For example, policy benefits may boost political participation by incentivizing individuals to mobilize. Interpretive effects shape public attitudes through policy messaging, design, and implementation (Mettler and SoRelle 2018). These interpretive dimensions affect engagement by framing how policies are perceived (Martin 2004; Mettler and Soss 2004).

In personal credit systems, resource effects manifest through credit-checking services and material benefits tied to higher scores, encouraging active engagement. Although these dynamics exist broadly, they are expected to be stronger in cities with a local personal credit system, where incentives become more visible and directly tied to everyday interaction. Interpretive effects concern how the SCS, as a broader governance strategy, is communicated. Local implementation is likely to increase the visibility and salience of such messaging, reinforcing knowledge of credit assessments and normalizing system use. Consequently, residents in cities with a local SCS are expected to have greater awareness of the government SCS and more frequently monitor their credit records. The Chinese government uses various methods, including policy documents, interpretations, and role model narratives (Chen et al. 2022), to disseminate SCS goals and functions. This framing, along with the material benefits linked to credit scores, contributes to public support for the SCS as shown by Xu et al. (2025). This shift in attitudes can, in turn, indirectly influence public engagement with the SCS. Based on these policy theories and empirical findings, we developed Hypothesis 2:

H2. *Administrative system focus. Compared to citizens in cities without a local credit scoring system (C2), those in cities with such a system (C1) demonstrate higher overall engagement (knowledge, information seeking, and use of score for benefits) that is predominantly confined to the administrative credit system. Engagement* *in commercial and financial credit systems is not expected to differ significantly between C1 and C2 participants.*

3.3 | Variations Between Socio-Demographic Characteristics

This section develops hypotheses concerning the role of key individual socio-demographic characteristics, including age, gender, and socio-economic status, as predictors of engagement across the three credit systems in China. Research on financial literacy has demonstrated that financial behaviors, attitudes, and expectations are shaped by socio-demographic characteristics, including gender, age, education, and income (Borden et al. 2008; Hira et al. 1993). Specifically, while financial literacy is measured in different ways, research indicates that men generally exhibit higher levels of financial knowledge than women (Chen and Volpe 1998; de Bassa Scheresberg 2013; Lusardi and Mitchell 2008; Micomonaco 2003). Regarding age, the literature suggests a U-shaped relationship between age and financial literacy, peaking around age 53 (Agarwal et al. 2009). Additionally, higher income and educational levels are linked to greater financial literacy and more responsible behaviors (Bernerth 2012; Grohmann 2018; Robb and Woodyard 2011). Given that a key goal of the personal credit system is to underpin trustworthy financial decision-making, these empirical findings provide a foundation for understanding how socio-demographic characteristics influence engagement with personal credit systems. Building on this theoretical background, we hypothesize:

H3. *Gender. Male citizens are more engaged with all personal credit systems than female citizens.*

H4. *Age. Citizens' engagement across all personal credit systems increases with age up to 60 years, then declines.*

H5. *Socio-economic status. There is a positive association between socio-economic variables (education and income) and engagement across all personal credit systems.*

4 | Methodology

4.1 | Primary Data Sources and Questionnaire Design

This study uses a comprehensive questionnaire to examine individuals' perspectives on the development of personal credit systems in China. An online survey, conducted in Chinese between June and August 2023, was facilitated by a Vienna-based survey company that recruited participants through local branches/partners in China. Responses were stored directly on our server, allowing real-time monitoring during data collection.

The sampling strategy accounted for age and gender distribution among internet users across Chinese cities. We selected 19 Chinese cities designated as "SCS Construction Model Cities," as residents from these cities are expected to be more engaged with the personal credit systems and possess greater knowledge about them. Of the selected cities, 10 had implemented a local SCS personal credit scoring system (C1) and 9 had not (C2). Survey questions focused on (1) public understanding of credit-related concepts; and (2) participants' knowledge and use of the different personal credit systems, including the SCS blacklist and redlist scheme, Zhima Credit, the local scoring system, as well as the PBoC credit report. Two versions of the survey were created, differing only in questions related to the local SCS. The survey for participants from C1 included questions about their knowledge, checking behaviors, and use of scores related to the specific local SCS, whereas the survey for participants from C2 did not. Participants received the version corresponding to their city of residence. Sampling quotas were created based on age (18+), gender, city, and city population, using data from *The 51st Statistical Report on China's Internet Development* (China Internet Network Information Center (CNNIC) 2023) and official city government sources. In addition to these core representativeness criteria, information on education level, income, and employment status was also collected to facilitate robustness checks. To ensure survey validity, we employed several quality-control measures: an attention-check question, pilot testing of the questionnaire with a small subsample, and response-time monitoring to exclude inattentive or excessively brief answers.

To reach our goal of 5000 valid responses, we conducted multiple survey rounds, assessing data quality based on attention checks, survey completion times, and duplicate detection. Recruiting older participants proved challenging, leading to relaxed quotas in the final data collection round.

We measured people's engagement in different personal credit systems from three perspectives. First, we examined people's knowledge of major credit systems in China (e.g., question ii in Table 1). Second, we investigated individuals' utilization of these credit systems, focusing on their own score checks and inquiries into others' records. We categorized "others" into three groups: close personal relationships (e.g., family, friends); close working or business relationships (e.g., colleagues, business partners); and unfamiliar individuals. Participants reported the frequency and purposes of these inquiries (e.g., questions iii and iv in Table 1). Third, we explored individuals' use of credit scores which are often linked to various benefits. We analyzed usage frequency to understand the reward mechanisms embedded in these systems. Additionally, we also assessed participants' understanding of credit-related concepts (e.g., questions i and v in Table 1).

The questionnaire consisted of a mix of rating-scale (Likert-type scale), multiple choice, and open-ended questions (see Table 1). Since "social credit" is a relatively new concept, the perception-related question was informed by insights gained from 15 preliminary interviews: 10 with Chinese residents from diverse backgrounds and 5 with SCS researchers. Their answers were coded into 10 themes, which were incorporated as survey answer options, with an additional text box for alternative explanations.

4.2 | Data Analyses

We cleaned the data by excluding participants that failed the attention check, completed the survey too quickly (Leiner 2019), or had duplicated reference numbers. This process resulted in 5538 valid responses, including 2268 from cities with an established

SCS personal credit scoring system (C1) and 3270 from cities without one (C2). Table A1 in Appendix A provides an overview of respondents' demographic characteristics.

To analyze open-ended responses on public perceptions of "credit" and "social credit," we employed unsupervised topic modeling using the Latent Dirichlet Allocation (LDA) algorithm (Blei et al. 2003). This method was chosen because it allows underlying semantic structures to emerge inductively from the data, without imposing predefined categories, making it suitable for exploring how citizens articulate abstract concepts such as "credit" and "social credit." Using LDA, each response was represented as a probabilistic mixture of topics, and each topic as a distribution over words, capturing the diversity of meanings across participants. Responses were pre-processed to remove noise. Blank responses, entries containing only numbers or English letters, and a set of common non-informative phrases (e.g., "无/None," "没有/No," "挺好的/pretty good," "不清楚/not clear") were excluded. Only meaningful Chinese text responses were retained for analysis. We tested multiple topic numbers (3–10 topics) and selected the final solutions based on interpretability and thematic coherence, thereby choosing the number of topics that produced distinct and meaningful clusters without excessive overlap or fragmentation. The resulting five-topic model for "credit" ($k=5$) and six-topic model for "social credit" ($k=6$) revealed clear and interpretable themes (see Tables 2 and 3). While LDA provides topic–document weights, our analysis focuses on the dominant topics and their associated keywords to qualitatively interpret respondents' conceptualizations.

For statistical analysis, we used ANOVA to examine differences in knowledge and usage across credit systems, followed by Tukey's Honest Significant Difference (HSD) test for multiple comparisons. Welch's t-test and paired t-test were conducted to compare citizens' engagement between city groups (C1 vs. C2) and between different systems. Additionally, regression analysis examined the relationships between engagement factors and socio-demographic characteristics. These results provide insights into the role of

TABLE 1 | Sample questions and answers in our survey.

	Questions	Answers
Rating-scale questions	i. To what extent do you agree with the following explanation of "social credit"?: "Social credit" is an expectation of access to social resources.	1. Strongly disagree; 2. Disagree; 3. Neither agree nor disagree; 4. Agree; 5. Strongly agree
	ii. Please indicate to what extent you are familiar with Blacklists issued by the Social Credit System?	1. Never heard of; 2. Heard of, but not familiar; 3. Have some knowledge; 4. Fairly knowledgeable; 5. Very well-informed
	iii. How often did you check your Zhima Score during the past 12 months?	1. Never; 2. About once per year; 3. About 2–4 times per year; 4. About 5–9 times per year; 5. At least 10 times per year
Multiple-choice questions	iv. For what purposes do you search for information about individuals with whom you have close working relationships or business transactions on the Social Credit System's Redlist? (select all that apply)	1. To promote economic activities (e.g., online and offline transactions/buying and selling, apartment renting, job hunting, etc.); 2. To promote social activities (e.g., dating, social networking, etc.); 3. Out of curiosity
Open-ended questions	v. (Following the previous rating-scale questions) Other explanations about "social credit" (please describe)	A text box is offered.

TABLE 2 | Labels from LDA results of participants' understanding of "credit."

	Labels	Clues
Topic 1	Morality and integrity	Integrity, personal honesty, morality, ethical domain, referring to trust in life area
Topic 2	Trust and reputation	Trust among individuals, trust, credibility and reputation, moral trust, credit is trust
Topic 3	Interpersonal relationship	Interpersonal relationship, between people, trust between people, trust in interpersonal relationship, credit referring to trust among individuals
Topic 4	Public credibility	Politics, social credibility, society, public credibility
Topic 5	Credit and trust in economic and political contexts	Political domain, credit card, economy

personal credit systems across different population groups and their potential impact on financial inclusion.

Despite our efforts to ensure data quality through quota sampling, attention checks and real time recording, certain limitations remain. First, the survey's online format may bias participation toward digitally literate and urban populations, potentially underrepresenting older or rural residents. Second, responses regarding politically sensitive systems like the SCS may be subject to social desirability bias. While anonymity was ensured, respondents may still underreport concerns or negative experiences. Third, while this study was not formally preregistered, as it was not an institutional requirement at the time of data collection, we mitigated analytic flexibility by grounding our hypotheses in policy feedback theory prior to analysis (see Section 3). To ensure the robustness of our results, we employed theory-driven hypothesis testing alongside multiple regression specifications and comparative sub-sample analyses to verify the consistency of our findings across different implementation contexts. Finally, as with all survey data, we must distinguish between actual usage of systems and self-reported usage, which may differ.

4.3 | Ethical Issues

Our institution does not mandate ethics approval for surveys conducted in non-medical studies. However, we adhered to standard ethical research practices, including transparently presenting study procedures, obtaining informed consent, allowing participants to withdraw at any time, and ensuring data anonymity. To further respect participants' privacy, we deliberately avoided questions requiring sensitive information, such as their credit scores.

5 | Results

5.1 | Public Understanding of "Credit" and "Social Credit"

The term "credit" is derived from the Latin word "credere," meaning "to trust or believe."[8] Our survey indicates that most participants associate "credit" (信用) with both social and economic dimensions, with 88% (including both "agree" and "strongly agree") emphasizing social trust and 81% highlighting economic trust (see Figure 1), though there is a slightly stronger emphasis on the social aspects of "credit." The findings from topic modeling offer a more nuanced understanding of public perceptions. As LDA assigns probabilistic weights to each topic across individual responses, we interpreted the dominant topic distributions to identify overarching conceptual patterns while recognizing that individual perceptions may blend multiple themes. As shown in Table 2, the social attributes of "credit" are primarily explained through morality and integrity, trust and reputation, and interpersonal relationships (Topics 1, 2,

TABLE 3 | Labels from LDA results of participants' understanding of "social credit."

	Labels	Clues
Topic 1	About economic practice and order	Maintaining normal economic order, ethical and business conventions, benefiting the trustworthy, making the untrustworthy pay the price
Topic 2	Personal integrity	Honest and trustworthy, personal credit, personal integrity
Topic 3	Complement to laws and basis for social development	Promoting social progress, social credit is the foundation for social development, social credit is the complement to and guarantee of laws
Topic 4	Digital technology and big data	Artificial intelligence and block chain, such as big data
Topic 5	On the basis of laws and promoting a culture of trustworthiness	Based on relatively comprehensive legal system, based on legal system, enhancing the credit awareness of society members, establishing a trustworthy social environment
Topic 6	Basis for social norm, social stability, and interpersonal relationship	Basis for interpersonal relations, basis for social stability, basis for social norm, promoting social development

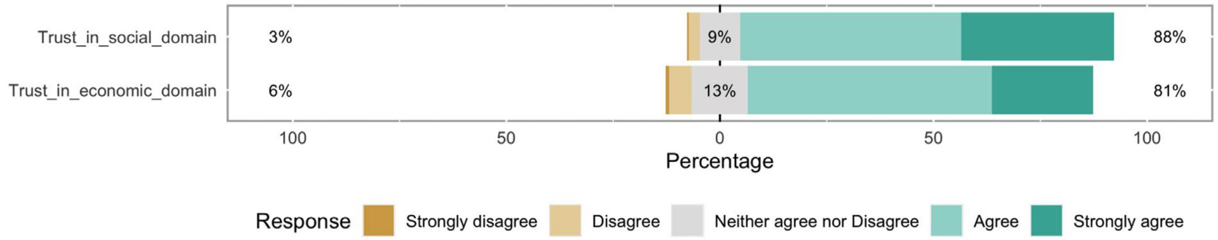

FIGURE 1 | Public understanding of "credit." Percentages on the left represent the sum of "strongly disagree" and "disagree" responses, and percentages on the right represent the sum of "strongly agree" and "agree" responses.

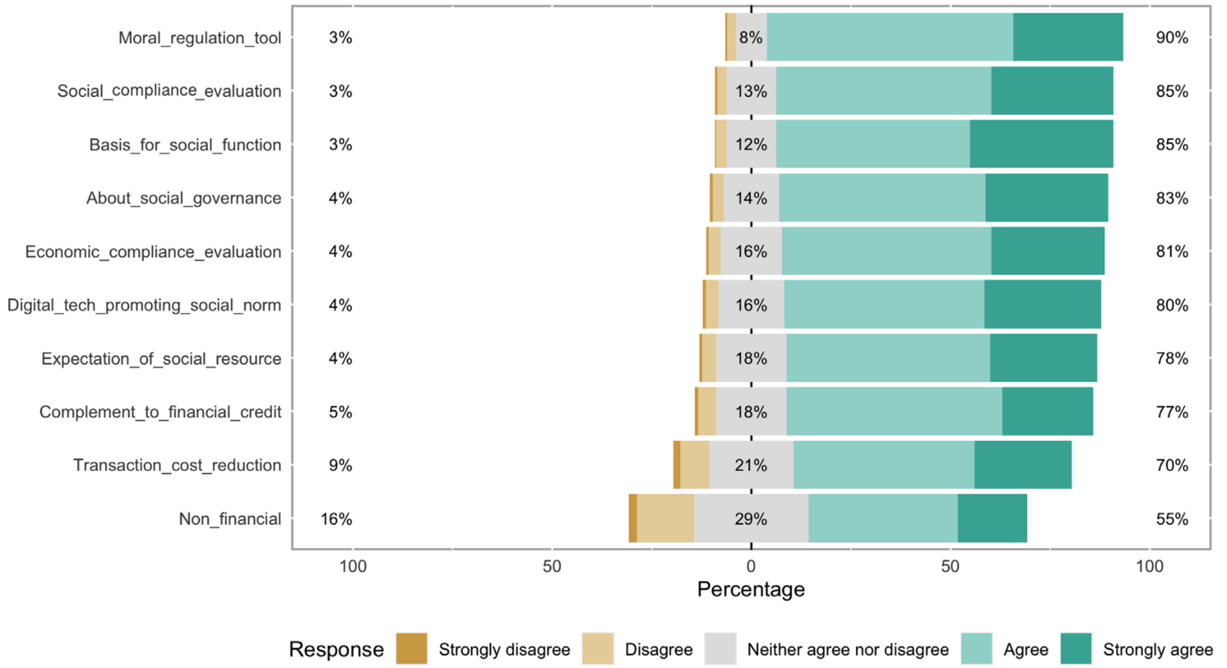

FIGURE 2 | Public understanding of "social credit." Percentages on the left represent the sum of "strongly disagree" and "disagree" responses, and percentages on the right represent the sum of "strongly agree" and "agree" responses.

and 3). This suggests that "credit" is regarded as a fundamental personal quality, playing a crucial role in social interactions. To account for conceptual overlaps, we compared the top keywords and exemplar responses across topics. While some terms—such as integrity, trust, and reputation—appeared in more than one topic, their contextual framing differed. For instance, one cluster emphasized morality and personal virtue, whereas another focused on interpersonal reliability and social reputation. This distinction suggests that participants associated "credit" with both ethical character and social relationships, but in meaningfully different ways. Additionally, participants distinguished between "credit" as public credibility and social trust (Topic 4) and "credit" as economic and political trust (see Table 2).

The concept of "social credit" is complex. Most participants (about 90%, including both "agree" and "strongly agree") correlated it with moral regulation (道德规范), while only 3% disagreed (including both "disagree" and "strongly disagree") and 8% were neutral (see Figure 2). Similar to "credit," participants linked "social credit" to both social and economic compliance, with a stronger emphasis on the social aspect (85% vs. 81%). Additionally, 77% saw "social credit" as complementing financial credit, while 55% considered it a non-financial concept. A majority (70%) acknowledged the system's role in reducing transaction costs, as highlighted in the *Planning Outline*, and 78% connected "social credit" to access to social resources (社会资源). "Social credit" was also linked to governance and societal functioning (社会治理和社会运作). Additionally, the concept was associated with leveraging digital technology to reinforce social norms (社会规范) (80%).

In the topic model, each theme contributes probabilistic weights to individual responses, forming a composite representation of how participants articulate the meaning of "social credit." This approach reveals that while morality, integrity, and trust frequently co-occurred with governance and technology-related themes, their contextual emphasis differed (see Table 3). For instance, responses that linked morality to digital oversight often portrayed technology as an enabler of social discipline, whereas those emphasizing law and order viewed it as an instrument for fairness and accountability. These overlaps indicate that citizens understand "social credit" not only as a moral framework but also as a digitally mediated governance tool. Overall, perceptions of "social credit" share conceptual ground with "credit" in moral and social dimensions but extend further into institutional and technological domains. This hybrid understanding fuses ethical, legal, and algorithmic elements, suggesting that "social credit" is perceived as both a moral compass and an infrastructural mechanism for shaping trustworthy behavior.

5.2 | Systems Dominance in Citizen Engagement

H1 states that citizens exhibit the highest overall engagement with commercial credit systems and the lowest with administrative credit systems, even in cities with a local administrative scoring system (C1). To test H1, we assessed citizen engagement with different credit systems across three dimensions: knowledge, information seeking, and use of credit scores for personal benefits. Engagement was compared across commercial, financial, and administrative systems.

5.2.1 | Knowledge of Credit Systems

We used knowledge of SCS blacklists and redlists (labeled as "the SCS lists") as an indicator of substantive understanding.

This aligns with research showing that citizens may be unfamiliar with the term "social credit system" but are aware of its key mechanisms (Chen and Grossklags 2022). We constructed a scale (1–5) based on the average responses to two lists, with a reliability analysis yielding a Cronbach's alpha of 0.72.

A repeated-measures ANOVA revealed significant within-subject differences in participants' knowledge across systems: $F(2,\ 16{,}608) = 140.5$, $p < 0.001$. Post-hoc paired t-tests (Bonferroni-corrected) confirmed that knowledge differed significantly between all systems ($p < 0.001$). Mean knowledge scores were highest for Zhima Credit ($M = 3.94$, $SD = 0.99$), followed by PBoC credit reports ($M = 3.68$, $SD = 0.97$) and SCS lists ($M = 3.19$, $SD = 1.00$) (see Figure 3). These patterns hold even in cities with local scoring systems (C1), where repeated-measures ANOVA and Bonferroni corrected pairwise comparisons further showed that participants were significantly less familiar with local credit scores than with the Zhima Score (diff = 1.41), PBoC reports (diff = 1.09), and SCS lists (diff = 0.65).

5.2.2 | Information-Seeking Behavior

Next, we measured participants' frequency of checking their own and others' credit records across systems (1–5 scale). We assessed the usage of national administrative systems for credit checks by averaging the frequency with which citizens check both SCS blacklists and redlists. Reliability was high for both scales (Cronbach's alpha of 0.9 for checking own records, 0.93 for checking others' records). We constructed one average scale for information seeking of *own* records, and one average scale for information seeking of *others'* records.

Repeated-measures ANOVA and post-hoc pairwise comparisons with Bonferroni correction revealed significant differences in participants' own credit check frequency across all three systems ($p < 0.001$). People most frequently seek information in Zhima Credit and least frequently in government-run systems. Specifically, 45.3% of participants checked their Zhima Score more than four times a year, compared to 16.1% for PBoC reports and 8.8% for SCS lists (see Figure 4). In C1, participants showed the same pattern: local credit scores and SCS lists were checked less frequently than the Zhima Score or PBoC credit reports.

When checking others' credit records, citizens engage less overall, but the pattern of system dominance persisted. A paired t-test revealed a statistically significant difference in checking frequency between Zhima Score and SCS lists ($p < 0.001$), suggesting that participants are more likely to consult others' Zhima Score than SCS list records. For example, only 3.8% of participants accessed SCS lists checking for others' credit information more than four times per year, with 65% doing so fewer than twice per year. While consultation of others' Zhima Score was also limited (51.4% checking fewer than twice per year), high-frequency usage (more than four times per year) was nearly double that of SCS lists (7% vs. 3.8%) (see Figure 5). In C1, this hierarchy remained consistent: the Zhima Score was accessed significantly more frequently than both SCS lists ($p < 0.001$) and local credit scores ($p < 0.001$), with the local scores being the least utilized source ($p < 0.001$).

5.2.3 | Use of Credit Scores for Personal Benefits

Across the full sample, approximately 70% of participants used the Zhima Score for services at least twice a year, and 24.32% reported using it 10 or more times annually. Since local credit scores are unavailable in C2, we compared the use of the Zhima Score and local credit scores among C1 participants. A paired t-test showed that the Zhima Score was used far more frequently than local credit scores ($p < 0.001$; see Figure 7). In C1, 25.09% used their Zhima score 10 or more times annually, whereas only 6.57% reported doing so for local credit scores.

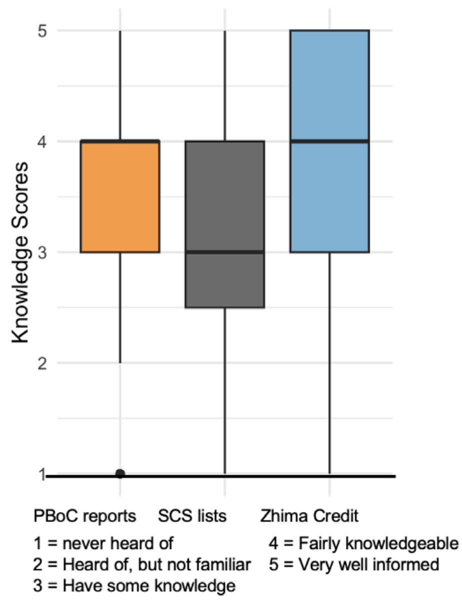

FIGURE 3 | Knowledge of credit systems.

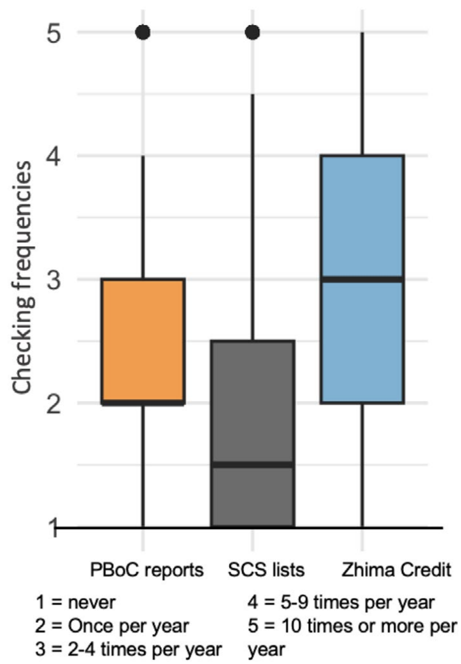

FIGURE 4 | Own credit records check.

In summary, across all three engagement dimensions (knowledge, information seeking, and use of credit scores), citizens exhibited highest engagement with commercial credit systems and lowest with administrative systems. This pattern held even in cities with a local credit scoring system (C1). These findings provide clear support for H1, demonstrating the dominance of commercial credit systems in China's personal credit landscape.

5.3 | Engagement by City Type

H2 states that, compared to citizens in cities without a local credit scoring system (C2), those in cities with such a system (C1) show higher overall engagement with administrative credit systems, while engagement with commercial and financial systems does not differ significantly. To test H2, we compared citizen engagement between cities with a local credit scoring system (C1) and those without such a system (C2). Again, engagement was examined across knowledge, information seeking, and use of credit scores for personal benefits.

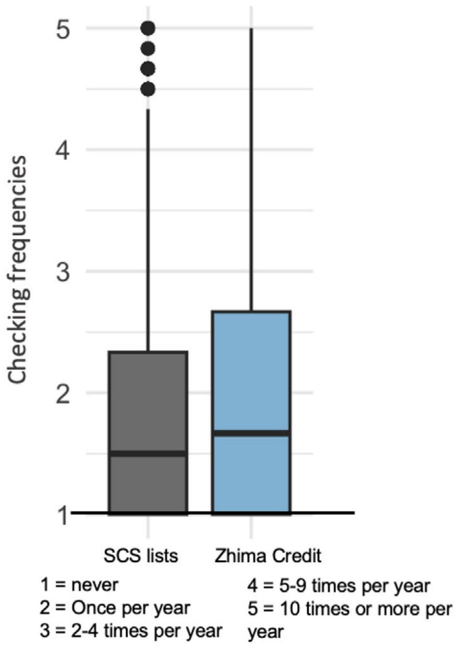

FIGURE 5 | Others' credit records check. Figures 3–5 utilize the full validated dataset ($N = 5538$). Each plot illustrates a different dimension of credit engagement.

5.3.1 | Knowledge of Credit Systems

Welch's t-tests suggested that C1 participants have marginally higher knowledge of the SCS lists than those in C2 (M: 3.25 vs. 3.15, $p < 0.001$). A comparable small difference was also observed for Zhima Credit knowledge (M: 4.00 vs. 3.90, $p < 0.001$). In contrast, knowledge of PBoC credit reports did not differ significantly between the two city types ($p > 0.1$).

5.3.2 | Information-Seeking behavior

Welch's t-tests show that C1 participants checked their own records in SCS lists more frequently than C2 participants ($p < 0.001$). However, no significant difference existed between C1 and C2 in checking others' SCS records ($p > 0.1$). A parallel pattern held for Zhima Score checks: C1 participants check their own Zhima Score more often than C2 participants ($p < 0.01$), while checks of others' Zhima Score do not differ significantly ($p > 0.1$). Checking frequency for one's own PBoC credit report did not differ significantly between C1 and C2 ($p > 0.1$).

To further explore credit-checking behaviors and purposes, we categorized target individuals into personal contacts (family/friends), work-related contacts (colleagues/business partners), and unfamiliar individuals. ANOVA results show significant differences in checking frequency across these three groups for both Zhima Score and SCS lists ($p < 0.001$). Paired t-tests indicate that credit records of unfamiliar individuals are checked far less frequently than those of personal contacts (Zhima Score: diff $= -0.72$; SCS lists: diff $= -0.43$) and than those of work-related contacts (Zhima Score: diff $= -0.69$; SCS lists: diff $= -0.53$). SCS lists are used slightly more frequently to assess work-related contacts than personal contacts (diff $= 0.10$, $p < 0.001$), whereas no corresponding difference was observed for the Zhima Score ($p > 0.1$).

Participants primarily reported checking others' credit records for economic purposes (e.g., renting, job hunting, and buying or selling goods online or offline), followed by social reasons

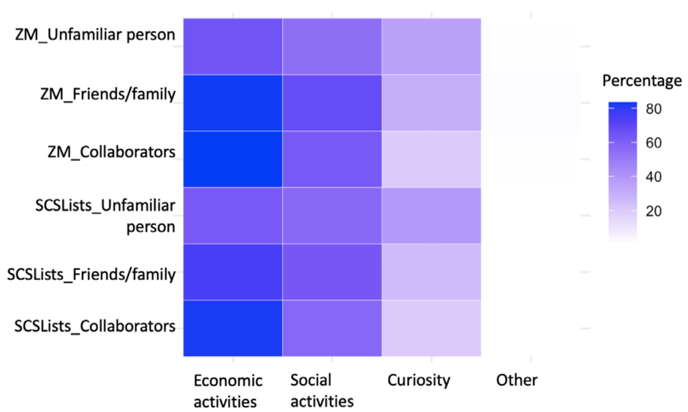

FIGURE 6 | Checking others' credit scores for different purposes.

(e.g., networking, dating). Curiosity-driven checking was comparatively rare (see Figure 6). This pattern suggests that while Chinese citizens emphasize the social dimensions of "credit" and "social credit," these attributes are often mobilized in economic contexts, where reputation functions as a useful and reliable indicator. This tendency aligns with prior research, which finds that credit scores are often embedded or required in well-defined economic scenarios (Wu et al. 2017).

5.3.3 | Use of Credit Scores for Personal Benefits

While local credit scores exist only in C1, comparisons between C1 and C2 groups remain applicable for the Zhima Score. However, no significant difference in Zhima Score usage was observed between C1 and C2 ($p > 0.1$).

Taken together, the results provide partial support for H2. As predicted, participants in C1 cities exhibited greater engagement with administrative credit systems, evidenced by higher knowledge of SCS lists and more frequent checking of their own SCS records. Engagement with financial credit, as measured by PBoC credit reports, did not differ across city types, consistent with our expectation. However, contrary to our expectations, participants in C1 cities also showed slightly higher knowledge of Zhima Credit and more frequent checking of their own Zhima Score compared to C2 participants. These differences, though statistically significant, were modest and likely reflect minor spillover effects in general credit awareness rather than a substantive increase in commercial credit engagement. Overall, these findings suggest that the introduction of local credit scoring systems primarily enhances administrative engagement. This pattern highlights the specific role of local systems in shaping citizen interaction with administrative credit mechanisms.

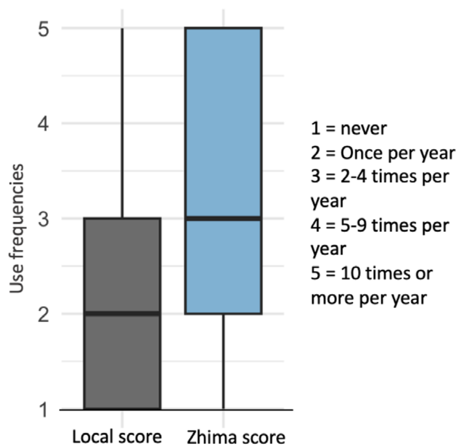

FIGURE 7 | Use of credit scores in C1.

5.4 | Socio-Demographic Factors

To evaluate H3–H5, we examined whether gender, age, and economic status were systematically associated with citizens' engagement with major personal credit systems. Regression analyses were conducted for three forms of engagement (see Tables A2 and A3).

5.4.1 | Gender

In a general sense, gender shows limited and inconsistent associations with engagement across personal credit systems. Female participants reported slightly lower knowledge of PBoC credit reports ($\beta = -0.06$, $p < 0.05$), while no significant gender differences emerged in knowledge of SCS lists or Zhima Credit (see Figure 8). Gender was also not systematically associated with checking one's own or others' credit records across the PBoC credit reports, SCS lists and Zhima Score (see Figures 9 and 10). There is also no association between gender and the use of the Zhima Score. In C1 cities, however, gender shows weak localized effects. Female gender was positively associated with knowledge ($\beta = 0.11$, $p < 0.05$), checking their own ($\beta = 0.14$, $p < 0.01$) as well as others' scores ($\beta = 0.09$, $p < 0.05$), and more frequent score usage ($\beta = 0.27$, $p < 0.001$). These effects appear specific to the local credit scoring systems and do not generalize to other systems. Overall, the pattern does not support H3, which predicted higher engagement among men across systems. Instead, gender differences are minimal for different systems at the national level, with modest female-leaning engagement emerging only in cities operating local credit scoring systems.

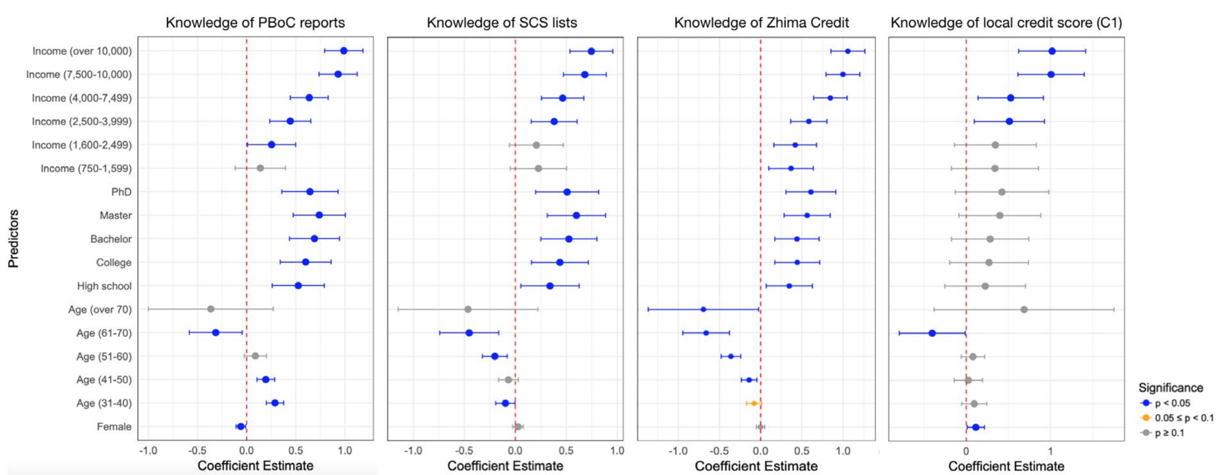

FIGURE 8 | Coefficient estimates of knowledge of different personal credit systems.

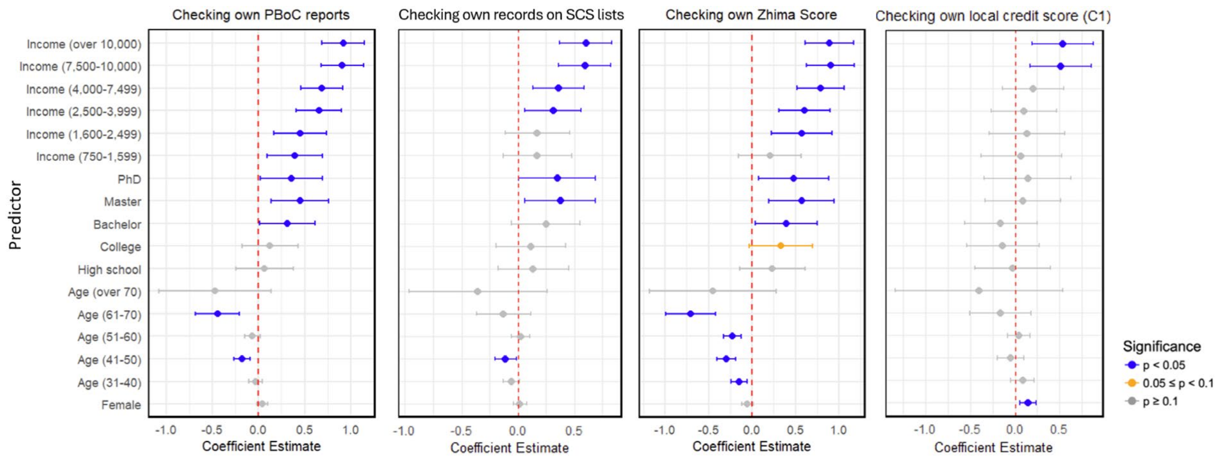

FIGURE 9 | Coefficient estimates of checking own records in different personal credit systems.

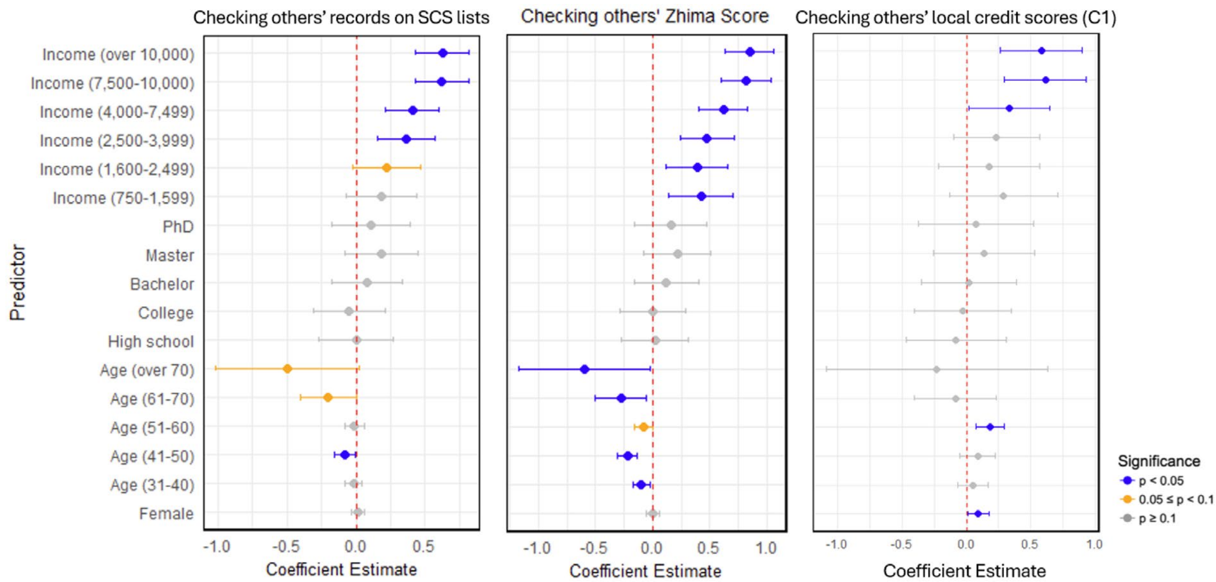

FIGURE 10 | Coefficient estimates of checking others' records in different personal credit systems.

5.4.2 | Age

Age effects were inconsistent across systems and engagement dimensions. Knowledge of PBoC credit reports increases from the 31–40 to the 51–60 age groups ($\beta = 0.27$, 0.19, 0.20, all $p < 0.001$), consistent with rising knowledge in mid-life, before declining sharply among those aged 61–70 ($\beta = -0.23$, $p < 0.05$) and over 70 (n.s.). For Zhima Credit, knowledge peaks in early adulthood and then declines earlier than expected: ages 41–50 ($\beta = -0.12$, $p < 0.01$), 51–60 ($\beta = -0.08$, $p < 0.05$), 61–70 ($\beta = -0.43$, $p < 0.001$), and over 70 ($\beta = -0.69$, $p < 0.05$) all show progressively lower knowledge. For SCS lists, age does not show the predicted increase during mid-life; coefficients for ages 31–60 are small and non-significant. However, knowledge declines substantially in the 61–70 group ($\beta = -0.26$, $p < 0.05$), suggesting that only the later-life decrease is evident. For local credit scores in C1 cities, no mid-life increases appear, but a clear decline emerges among ages 61–70 ($\beta = -0.40$, $p < 0.05$).

Patterns of information seeking show a different age trajectory. In terms of checking own credit records in PBoC reports and SCS lists, no consistent mid-life pattern appears. Instead, the first consistent decline appears only among the 41–50 group for PBoC credit reports ($\beta = -0.18$, $p < 0.001$) and again at 61–70 ($\beta = -0.44$, $p < 0.001$). For checking the own Zhima Score, reductions begin earlier and grow steeper: ages 31–40 ($\beta = -0.15$, $p < 0.01$), 41–50 ($\beta = -0.30$, $p < 0.001$), 51–60 ($\beta = -0.23$, $p < 0.001$), and 61–70 ($\beta = -0.70$, $p < 0.001$). For local credit scores in C1, no mid-life differences appear, but those aged 61–70 check their own scores less often ($\beta = -0.17$, n.s.). In terms of checking others' credit records, a similar declining trend appears for the Zhima Score, with significant decreases beginning at 31–40 ($\beta = -0.10$, $p < 0.01$) and intensifying through older age groups. For other systems, only scattered age differences emerge (e.g., SCS lists at 41–50: $\beta = -0.08$, $p < 0.05$).

Use of Zhima Score (for personal benefits) is strongly negatively associated with age: 31–40 ($\beta = -0.34$), 41–50 ($\beta = -0.66$), 51–60 ($\beta = -0.53$), 61–70 ($\beta = -1.00$, all $p < 0.001$), and > 70 ($\beta = -1.29$, $p < 0.01$, see Figure 11). The pattern is a clear monotonic decline in score usage with age. Use of local credit scores

(C1) also declines at older ages, but these associations are not statistically significant.

Taken together, these results provide partial support for H4. Engagement does not consistently increase through mid-life across all systems, but there is robust evidence of declining engagement after age 60, especially for commercial credit systems. The expected mid-life peak appears clearly only for knowledge of PBoC credit reports. In general, age gradients are more pronounced for information seeking than for knowledge, and more consistent for commercial than administrative credit systems.

5.4.3 | Socio-Economic Status (Education and Income)

To assess the role of socio-economic status in credit system engagement (H5), we examined two indicators—education and income as predictors of knowledge, information seeking, and use of credit scores for personal benefits across all credit systems.

5.4.3.1 | Education. Higher educational achievement is consistently associated with greater knowledge of PBoC credit reports, SCS lists, and Zhima Credit. Specifically, relative to participants with middle-school education or below (reference group), those with high school education and above display large, monotonic increases in knowledge of these three systems. Effect sizes are sizable and statistically significant for nearly all education categories (e.g., bachelor's degree: $\beta = 0.69$ for PBoC credit reports, $\beta = 0.55$ for SCS lists, $\beta = 0.47$ for Zhima Credit, all $p < 0.001$). However, this pattern does not hold for knowledge of local credit scoring systems in C1 cities where educational differences are not statistically significant. For information seeking, education is positively associated with checking one's own Zhima Score and PBoC credit reports at bachelor's level and above (e.g., master's: $\beta = 0.57$, $p < 0.01$), but associations are weaker or insignificant for checking SCS lists and for local credit score checks among C1 participants. Checking others' credit records does not exhibit systematic education-related differences across any systems. Education also predicts use of credit scores for personal benefits, particularly for Zhima Score (e.g., master's degree: $\beta = 0.78$; PhD: $\beta = 0.73$, both $p < 0.001$). In C1 cities, this pattern exists only among participants with a PhD degree ($\beta = 0.77$, $p < 0.01$).

5.4.3.2 | Income. Income exhibits a similarly strong, gradual pattern. Compared with the lowest income category (below 750 yuan/month), higher income groups (at and above 2500 yuan) show significantly higher knowledge across all systems. For knowledge of PBoC credit reports, SCS lists, and Zhima Credit, effects grow from moderate (2500–3999 yuan, β: 0.36–0.56) to high income brackets (e.g., over 10,000 yuan: $\beta = 0.94$ for PBoC reports, $\beta = 0.72$ for SCS lists, $\beta = 1.01$ for Zhima Credit; all $p < 0.001$). Similar patterns appear for knowledge of local credit scores in C1 cities, where the highest income groups show substantial increases ($\beta = 1.02$, $p < 0.001$). Income is positively and significantly associated with individuals' likelihood of checking both their own and others' credit records across systems. Compared with the lowest income group, individuals earning 2500–3999 yuan are more likely to check their own credit records (β: 0.31–0.66), with effects growing at higher income levels (e.g., over 10,000 yuan: β: 0.60–0.92, all $p < 0.001$). For checking others' records, significant increases appear from 2500 yuan and above (e.g., $\beta = 0.37$ for SCS lists, $\beta = 0.48$ for Zhima Score). Local credit score checks for both one's own and others' scores in C1 cities show notable effects from 4000 yuan. Income also strongly predicts use of credit scores. For Zhima Score use, all income categories above 1600 yuan show large, significant increases (e.g., 4000–7499 yuan: $\beta = 0.85$, $p < 0.001$; over 10,000 yuan: $\beta = 1.20$, $p < 0.001$). Use of the local credit score in C1 cities follows the same gradient,

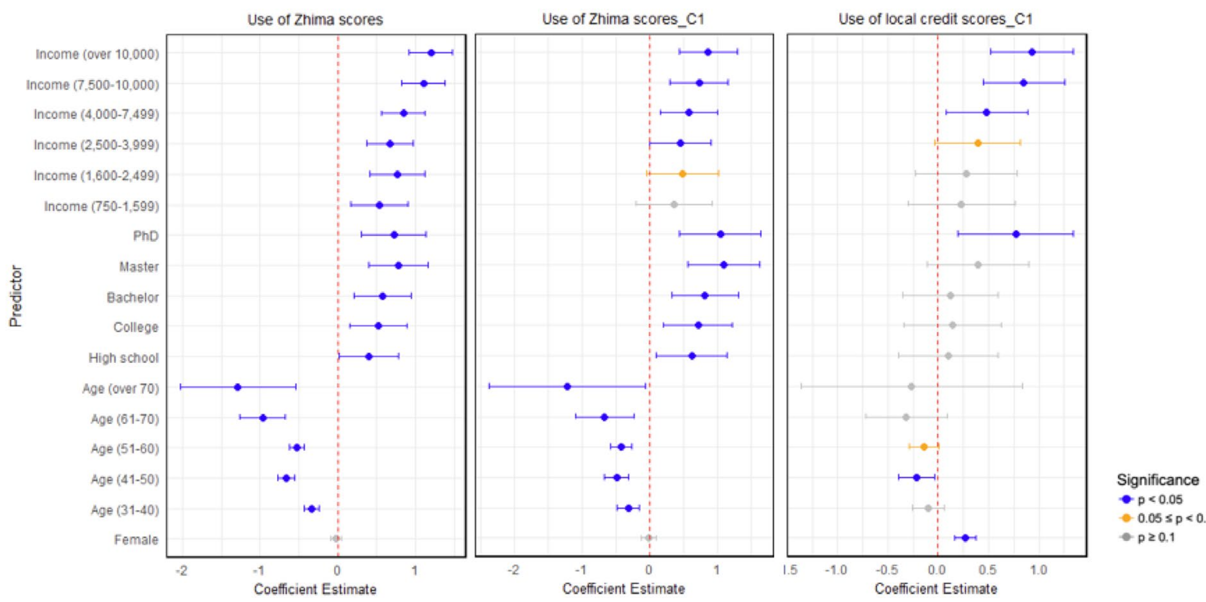

FIGURE 11 | Coefficient estimates of use of different personal credit scores.

with the highest income group reporting the largest effects ($\beta = 0.93$, $p < 0.001$).

Together, education and income show strong, consistent, and gradually increasing effects on knowledge, information seeking, and use of credit scores across personal credit systems. These findings provide clear support for H5: socio-economic status seems to be a major determinant of engagement with China's personal credit infrastructure. Highly educated and higher-income individuals exhibit higher knowledge, more intensive use, and somewhat greater information-seeking, especially for commercial credit scores (Zhima) and, within C1 cities, local credit scores.

6 | Discussion

In this section, we situate the findings within theoretical frameworks of algorithmic governance, policy feedback, governmentality, and social sorting, offering insights into their implications for digital inequality, financial inclusion, and the socio-political dynamics of data-driven governance in non-Western contexts.

6.1 | Understanding "Credit" and "Social Credit" in a Non-Western Context

The *Cambridge Dictionary* defines "credit" as having dual meanings. In the *financial* domain, it is "a method of paying for goods or services at a later time," and in the *social* domain, it relates to "praise, approval, or honour,"[9] which connects to social norm. Although "credit" and "social credit" have been widely discussed in Chinese policy and academic discourse, there remains a lack of systematic research on how the public interprets these concepts. Existing studies primarily examine the institutional design and governance logic of the SCS (e.g., Creemers 2018; Shen 2019) rather than the broader conceptual or cultural meanings of the concepts themselves. Initial evidence from prior survey-based studies indicates that the Chinese public's understanding links "credit" to moral behavior and social stability within the broader SCS framework (Kostka 2019; Liu 2022).

The rapid development of personal credit systems in China has been closely guided by, and integrated into, the broader SCS governance framework, shaping how Chinese residents perceive "credit" and "social credit." For most participants, "social credit" is not an abstract concept but synonymous with the institutionalized social credit system framework. As noted earlier, SCSs evaluate individuals' trustworthiness based on both economic and social behaviors, extending their influence beyond financial transactions into social governance. For instance, in Shanghai, improper waste sorting is classified as "bad" behavior (Shanghai City Management Administration Law-enforcement Bureau 2019), and individuals labeled as dishonest and subject to enforcement (referred to as "Lao Lai") may be barred from enrolling their children in private schools. Conversely, those with high local credit scores may enjoy benefits like deposit-free library services. While 16% of participants contested the notion that "social credit" is non-financial, a majority (55%) perceived it as a non-financial concept, and 77% regarded it as complementary to financial credit. Similarly, "credit" is more commonly perceived as a social rather than economic concept.

According to Foucault's concept of governmentality, modern governance shapes individuals' conduct through dispersed, non-coercive mechanisms (Foucault 1991). As an algorithmic regulation tool, the SCS extends governmentality by embedding state-defined notions of trustworthiness into everyday social and economic interactions. The strong association of "social credit" with moral regulation and digital technology suggests that citizens internalize the SCS as a mechanism that not only evaluates financial reliability but also constructs normative social behaviors. The internalization reflects a form of disciplinary power, where individuals self-regulate in response to state surveillance and incentives (Foucault 1977). The emphasis on digital technologies in public perceptions further underscores the role of algorithmic systems in rendering governmentality more pervasive, as automated scoring systems make surveillance and behavioral steering seamless and ubiquitous.

Additionally, the dominant themes around morality, integrity, and interpersonal relationships echo traditional Confucian values that remain embedded in contemporary Chinese social structures. China is widely recognized as a relationship-based society, where *guanxi* (interpersonal relationships) plays a central role in economic, political, and social activities (Buttery and Wong 1999). In this context, creditworthiness is closely tied to *guanxi* (Luo and Chen 1996), as an individual's credibility is often assessed based on the strength and quality of their social relationships (Luo 2007). Factors such as social reputation, educational achievements, past interactions, and the strength of one's *guanxi* network significantly shape perceived creditworthiness (Luo 2007). This intrinsic connection elucidates why participants in our study consistently emphasized *guanxi* when discussing both "credit" and "social credit," underscoring the association between credit and personal integrity. Consequently, "social credit" is increasingly perceived as a mechanism for moral regulation—one that is enabled and amplified by digital technologies and big data analytics (see Figure 2 and Table 3).

Further, China's socio-economic and cultural traditions prioritize harmony, stability, and familial values (Haiyan et al. 2024; Li 2006). Within this framework, political order, social norms, and interpersonal relationships fundamentally shape public conceptions of "credit" and "social credit." Participants associated "credit" with political attributes, framing it as *public credibility* and linking it to trust in both economic and political spheres (see Table 2). Additionally, they interpreted "social credit" as an institutional tool for maintaining social stability (see Table 3). This dual perception reflects how traditional values intersect with modern governance mechanisms to redefine trust in contemporary China. These responses illustrate that cultural values related to morality and hierarchy can serve as one interpretive frame for understanding the SCS and its associated digital governance mechanisms. In this sense, social credit is less a novel invention than a digital re-articulation of existing social expectations, which aligns with Foucault's notion of governmentality adapted to specific cultural milieus.

6.2 | Engagement Patterns Across Systems

Our findings show that citizen engagement with personal credit systems in China is not the same but follows certain patterns across the three types of systems: commercial, financial, and administrative. Across all three dimensions of engagement, knowledge, information seeking, and use of credit scores for benefit, the commercial system consistently occupies a dominant position.

Participants reported the highest level of knowledge for the commercial system, followed by the financial system, while the administrative ones, in particular local scoring ones, were least understood. This pattern persists even in cities where a local scoring system is implemented. The same gradient is reflected in information-seeking behavior. Participants reported turning most often to commercial systems for checking their own and others' credit records. Information seeking through the financial system occurred less frequently, while national and local administrative systems were accessed least often. This pattern repeats when looking at the use of credit scores for benefits. Commercial credit scores were most frequently used to access tangible advantages such as discounted services, streamlined access to rentals, transportation convenience, or leisure benefits. Although local administrative systems also provide reward mechanisms, they were used only marginally, with nearly half of respondents in relevant cities reporting never having used the local score. However, even though the local administrative scoring system has limited implications for engaging with the system itself, its presence in a city appears associated with broader engagement with the personal credit landscape. People living in a city with a local scoring system reported higher knowledge of both commercial and administrative systems, and more frequent information seeking than people in cities without such a system.

These patterns have implications for regulation and governance. The prominence of the commercial credit system reflects how credit infrastructures in China become embedded in everyday digital practices such as mobile payments, online consumption, and platform-based services. Engagement appears shaped by convenience, visibility, and seamless integration rather than formal obligation, which is consistent with research on platform governance and digital behavioral steering (Li et al. 2025; Yeung 2018). From a governmentality perspective, such infrastructures may contribute to normalizing the idea that personal behavior can be evaluated and scored, as credit scores become routine features of digital participation. At the same time, the association between the presence of a local administrative scoring system and broader engagement implies that administrative infrastructures may exert effects indirectly, through visibility, interpretive cues, and signaling institutional salience (Pierson 1993). Even though use remains limited, the mere presence of these systems signals institutional importance and conveys expectations of responsible citizenship, aligning with analyses of the SCS as a tool shaping behavioral norms and civic identity (Creemers 2018). In this way, administrative credit systems may contribute to institutionalizing the notion that personal records can have social consequences, even without frequent direct engagement.

Taken together, these findings suggest a differentiated but potentially complementary ecosystem. The commercial system embeds credit practices in routine consumption, the financial system sustains formal and institutionally recognized accountability mechanisms, and administrative systems appear to signal a broader cultural and regulatory environment in which evaluative infrastructures matter. Rather than functioning as competing models, these systems appear to work in parallel, contributing to the normalization of credit-based governance in daily life.

6.3 | Disproportionate Benefits in Financial and Economic Inclusion Promoted by Personal Credit Systems in China

The World Bank defines financial inclusion as follows: "individuals and businesses have access to useful and affordable financial products and services that meet their needs – transactions, payments, savings, credit and insurance – delivered in a responsible and sustainable way" (World Bank Group 2024). By facilitating credit expansion, financial inclusion allows individuals to increase borrowing, consumption, and investment, thereby stimulating demand and investment. Crucially, access to credit serves as an economic equalizer, empowering disadvantaged groups to seize opportunities that would otherwise be out of reach (Han and Melecky 2013). This function positions financial inclusion as a key catalyst for economic empowerment and reduced income inequality. Recognizing its transformative potential, financial inclusion directly contributes to achieving 7 of the 17 Sustainable Development Goals, underscoring its role as a cornerstone of inclusive growth.

As previously discussed, advancing financial inclusion constitutes a critical component of the SCS's overarching objective to stimulate high-level economic development. This focus holds particular importance in China for three reasons. First, although the PBoC facilitates the world's largest personal credit reporting system, approximately 262 million citizens remain excluded from its coverage.[10] Second, China's Gini coefficient has remained consistently high, fluctuating between 0.46 and 0.47 (Hu and Cheng 2022), underscoring significant wealth disparities. Third, while formal bank financing has grown steadily over the past decade, informal financial channels – often relying on reputation and interpersonal relationships – continue to dominate China's financial landscape (Allen et al. 2005). Notably, despite the continuous growth of formal financing from banks during the past decade, informal financing has consistently surpassed the popularity of formal financing in China (Liu and Yin 2024). Formal financial markets predominantly serve borrowers with established credit histories and high credit scores (Garmaise 2015), whereas informal financial institutions primarily cater to underserved market segments, including small business owners, rural households, and low-income individuals (Ayyagari et al. 2010). The SCS aims to address this by expanding formal financial market access, prioritizing financial inclusion for vulnerable populations, and ultimately stimulating economic growth while reducing inequality (Engelmann et al. 2019; State Council 2024).

In our research, individuals' knowledge of personal credit systems, their credit information seeking behavior, as well as their use of credit scores for accessing benefits constitutes active financial market engagement. This behavior reflects personal financial management, preparation for loans or major purchases, and direct involvement in economic and social activities. While our cross-sectional data cannot establish causal relationships, it allows us to identify patterned associations between socioeconomic characteristics and engagement with personal credit systems. We found that income is a significant predictor of credit system usage, particularly for individuals earning over 2500 yuan (about 343 USD) per month, with effect sizes strengthening at higher income levels (see Table A3). Higher-income individuals show increased credit system utilization and greater financial market engagement. In contrast, no comparable pattern exists among lower-income groups. In China, approximately 547 million people (39.1%) earn less than 1000 yuan (about 138 USD) per month, and 964 million people (68.9%) earn less than 2000 yuan (about 275 USD) (Hu and Cheng 2022). These descriptive patterns suggest a potential stratification effect, where individuals with higher income and education are more likely to benefit from personal credit systems, while economically disadvantaged groups remain underrepresented. This reveals a systemic imbalance where a wealthier minority disproportionately benefits from personal credit systems and the economically disadvantaged majority experiences limited financial inclusion. However, we emphasize that these findings reflect associations rather than direct causal effects. Further longitudinal or experimental research is needed to assess whether and how engagement in credit systems causally contributes to economic inequality.

Furthermore, disparities raised by China's personal credit systems exemplify data colonialism, as described by Couldry and Mejias, wherein human behaviors and social interactions are extracted and commodified under the guise of economic optimization (Couldry and Mejias 2019). Concurrently, these systems embody Lyon's concept of social sorting, where algorithmic assessments categorize and stratify individuals, often amplifying existing inequalities by favoring those with robust financial and digital profiles (Lyon 2003). Rather than asserting that these systems directly produce inequality, our findings illustrate how their design and usage patterns may interact with pre-existing socio-economic structures, contributing to unequal digital and financial engagement. Far from merely reflecting financial status, these systems actively construct new forms of digital stratification, reinforcing economic privilege through algorithmic regulation and perpetuating socio-economic disparities.

7 | Conclusion

This study provides a comprehensive examination of how Chinese citizens perceive and engage with personal credit systems, including the financial credit reporting system, commercial credit systems, and the administrative SCS. By integrating a large-scale survey ($N = 5538$) with statistical and text analyses, our findings reveal nuanced public perceptions of "credit" and "social credit" and highlight significant variations in engagement across systems and socio-demographic groups.

A key contribution of this study lies in distinguishing between the three types of personal credit systems and empirically documenting their differential uptake and perceived functions. By disentangling the engagement with these systems, we provide a clearer understanding of how digital infrastructures are experienced and navigated in everyday life, challenging oversimplified narratives about China's SCS.

Our analysis also reveals the dual framing of "credit" and "social credit" as both economic and moral constructs, shaped by local cultural norms and state-driven narratives. Additionally, both the multi-option survey and topic modeling reveal strong associations between "social credit" and digital technologies. Chinese citizens exhibit notably optimistic views toward science and technology (Li et al. 2015; WARC 2018; Zhang 1991), with consistently positive attitudes spanning generations and extending across domains such as education, agriculture, and governance (Chen 2023; Huang et al. 2020; Li et al. 2019; Su et al. 2022; Wu and Qiu 2024). This widespread digital optimism—coupled with China's globally low proportion of "digital doubters" (Kostka 2023)—likely contributes to favorable perceptions of the SCS. However, such support must be interpreted with caution. In authoritarian contexts, public endorsement often reflects alignment with state-framed narratives rather than fully informed, voluntary consent (Chen et al. 2022; Creemers 2018). Engagement with the SCS may thus result as much from the political environment and limited space for dissent as from genuine public approval.

Our findings reveal notable disparities in public engagement across the credit systems that can be interpreted through the lens of policy feedback theory (Mettler and Soss 2004), which posits that public policies shape citizens' behaviors and attitudes through resource and interpretive effects. The high engagement with Zhima Credit reflects strong resource effects, as its integration into a wide range of everyday services (e.g., healthcare, shopping, apartment rental) provides tangible benefits that incentivize engagement. In contrast, the lower engagement with SCS lists and local credit scores may stem from weaker resource effects, as these systems offer less score/records application scenarios. Additionally, regional variations between the C1 and C2 groups reflect how local institutional arrangements mediate citizens' exposure to algorithmic governance. As our framework suggests, local policy implementation acts as a feedback mechanism that shapes engagement. Cities with stronger digital infrastructures and more proactive governance cultures appear to foster greater knowledge and use of credit systems, illustrating how contextual feedback reinforces engagement patterns. Income consistently emerged as the most influential socio-demographic factor shaping credit system engagement. Wealthier individuals are more familiar with these systems, use them more frequently, and derive greater personal benefits—an outcome that reinforces existing socio-economic hierarchies.

This supports the idea that those with more resources are better positioned to capitalize on institutional tools, creating a feedback loop that further entrenches inequality (Pierson 1993).

Beyond financial inequality, China's increasing dependence on big data analytics, AI-powered credit scoring, and alternative data sources for creditworthiness evaluation raises concerns about algorithmic bias, lack of transparency, and the broader implications of data-driven governance (Sadok et al. 2022). China's personal credit systems represent a transformation in social governance, where traditional credit mechanisms are evolving into tools for behavioral control. Data-driven assessments now influence not only loan approvals but also broader aspects of social mobility. As Pasquale criticizes with his concept of the "black box" (Pasquale 2015), scoring systems use opaque mechanisms to shape individuals' economic mobility and digital identities. Commercial credit systems, such as Zhima Credit, incorporate alternative data sources, such as online shopping habits, digital transactions, and social behaviors, to assess creditworthiness. As a result, low-income individuals with limited digital footprints may receive lower scores, restricting their access to credit and reinforcing financial exclusion. This reflects broader concerns about algorithmic bias and digital inequality in credit markets, where opaque data-driven decision-making systems often replicate existing socio-economic hierarchies rather than mitigating them (Lutz 2019; van Deursen and van Dijk 2014). Consequently, this exacerbates wealth inequality, contradicting the original goal of these systems to promote financial inclusion and equal opportunity, particularly for marginalized socio-economic groups.

The transmission of policy to the economy and society is widely acknowledged to exhibit long and variable lags (Dey et al. 2021; Ding et al. 2020; Havranek and Rusnak 2013). Given the complexity and scale of personal credit systems in China, these delays in response and impact are likely even more prolonged. As global reliance on AI-powered financial infrastructures grows, the Chinese case offers critical insights into the broader challenges of digital citizenship, financial inclusion, and algorithmic power. Future research should continuously monitor and assess the real-world impacts of these systems, ensuring that big data infrastructures do not systematically disadvantage vulnerable populations. Moreover, further research should investigate how regulatory frameworks—both in China and internationally—can enhance algorithmic accountability, prevent discriminatory credit practices, and address the long-term implications of financial datafication. Without greater transparency and safeguards against data-driven discrimination, digital credit systems risk exacerbating, rather than mitigating, social and financial inequalities.

Finally, while rooted in the Chinese context, our findings resonate with global trends in algorithmic governance. Systems such as Aadhaar (in India), the U.S. Department of Government Efficiency (DOGE), and the discontinued SyRI in Netherlands raise similar concerns about data justice, algorithmic opacity, and socio-economic stratification. China's case offers a unique, large-scale example of integrated digital governance, but it also serves as a cautionary tale of how these systems can deepen existing inequalities if not carefully regulated. Comparative research on these infrastructures can enrich our understanding of the global political economy of digital regulation.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Endnotes

[1] Please review the report "What Worries the World" from Ipsos between 2016 and 2019 at https://www.ipsos.com/en. China has not been included in the Ipsos survey since 2020.

[2] http://www.pbc.gov.cn/redianzhuanti/118742/4657542/4677461/index.html (in Chinese).

[3] Qiantang Credit gained the license in November 2024, after the survey was conducted for this research. Additionally, the establishment of a fourth personal credit agency is in process.

[4] Refer to the companies' official websites: https://www.baihangcredit.com, https://www.pudaocredit.cn.

[5] Information sourced from Alipay mobile application.

[6] AliPay and Zhima Credit are two interconnected products offered by Ant Financial. As of the end of June 2020, AliPay boasted over 1 billion annual active users, and the number kept growing. Based on this vast user base and the close relationship between AliPay and Zhima Credit, we estimate a similar number of users for Zhima Credit.

[7] Data source: http://www.pbc.gov.cn/redianzhuanti/118742/4657542/4677461/index.html and https://www.statista.com/statistics/1395691/china-alipay-monthly-active-users/.

[8] See https://www.merriam-webster.com/dictionary/credit.

[9] See https://dictionary.cambridge.org/dictionary/english/credit.

[10] This is calculated by the author as follows: 1.412 billion (total population of China) − 1.15 billion (that are covered by the PBoC personal credit reporting system) = 0.262 billion.

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Appendix A

Participants' Socio-Demographic Characteristics

TABLE A1 | Socio-demographic variables overview.

Variable	Category	n	%
Gender	Male	2705	48.84%
	Female	2827	51.05%
	Prefer not to say	6	0.11%
Age	18–30	1889	34.11%
	31–40	1629	29.41%
	41–50	792	14.3%
	51–60	1130	20.4%
	61–70	83	1.5%
	over 70	12	0.22%
	Prefer not to say	3	0.05%
Education	Middle school and below	52	0.94%
	High school (including vocational high school, technical secondary school, and technical college)	386	6.97%
	College degree	802	14.48%
	Bachelor's degree	3394	61.29%
	Master degree	686	12.39%
	Ph.D.	202	3.65%
	Prefer not to say	16	0.29%
Income (per month)	Less than 750 RMB (about 100 Euro)	93	1.68%
	750–1599 RMB	101	1.82%
	1600–2499 RMB	123	2.22%
	2500–3999 RMB	355	6.41%
	4000–7499 RMB	1512	27.3%
	7500–10,000 RMB	1394	25.17%
	More than 10,000 RMB	1860	33.59%
	Prefer not to say	100	1.81%

Regression Analysis

TABLE A2 | Regression analysis results (1).

Variables	Knowledge of major personal credit systems				Use of credit scores		
	PBoC reports	SCS lists	Zhima Credit	Local credit score (C1)	Zhima Score	Zhima Score (C1)	Local credit score (C1)
(Intercept)	2.21*** (0.16)	2.10*** (0.17)	2.66*** (0.17)	1.44*** (0.29)	2.02*** (0.22)	2.10*** (0.32)	1.18*** (0.30)
Gender (female)	−0.06* (0.02)	0.02 (0.03)	−0.01 (0.03)	0.11* (0.05)	−0.02 (0.04)	−0.00 (0.06)	0.27*** (0.05)
Age (31–40)	0.27*** (0.03)	−0.03 (0.04)	0.02 (0.03)	0.09 (0.08)	−0.34*** (0.05)	−0.31*** (0.08)	−0.10 (0.08)
Age (41–50)	0.19*** (0.04)	−0.08 (0.04)	−0.12** (0.04)	0.02 (0.09)	−0.66*** (0.06)	−0.48*** (0.09)	−0.22* (0.09)
Age (51–60)	0.2*** (0.04)	0.04 (0.04)	−0.08* (0.04)	0.08 (0.07)	−0.53*** (0.05)	−0.42*** (0.08)	−0.14 (0.07)
Age (61–70)	−0.23* (0.10)	−0.26* (0.11)	−0.43*** (0.11)	−0.40* (0.20)	−1.00*** (0.15)	−0.66** (0.22)	−0.32 (0.21)
Age (over 70)	−0.26 (0.26)	−0.34 (0.28)	−0.69* (0.28)	0.69 (0.54)	−1.29** (0.38)	−1.22* (0.59)	−0.26 (0.56)
Education (high school)	0.51*** (0.13)	0.33* (0.15)	0.35* (0.14)	0.22 (0.24)	0.41* (0.19)	0.63* (0.27)	0.10 (0.25)
Education (college)	0.59*** (0.13)	0.45** (0.14)	0.46*** (0.14)	0.27 (0.24)	0.53** (0.19)	0.72** (0.26)	0.14 (0.25)
Education (bachelor)	0.69*** (0.13)	0.55*** (0.14)	0.47*** (0.14)	0.28 (0.23)	0.57** (0.19)	0.83** (0.25)	0.12 (0.24)
Education (master)	0.74*** (0.13)	0.62*** (0.15)	0.59*** (0.14)	0.40 (0.25)	0.78*** (0.19)	1.10*** (0.27)	0.39 (0.26)
Education (PhD)	0.62*** (0.15)	0.53*** (0.16)	0.63*** (0.15)	0.42 (0.28)	0.73*** (0.21)	1.05*** (0.31)	0.77** (0.29)
Income (750–1599)	0.14 (0.13)	0.22 (0.14)	0.36* (0.14)	0.34 (0.26)	0.54** (0.19)	0.37 (0.29)	0.23 (0.27)
Income (1600–2499)	0.24 (0.12)	0.20 (0.14)	0.40** (0.13)	0.34 (0.25)	0.77*** (0.18)	0.49 (0.27)	0.28 (0.26)
Income (2500-3999)	0.42*** (0.11)	0.36** (0.12)	0.56*** (0.11)	0.51* (0.21)	0.68*** (0.15)	0.45* (0.23)	0.39 (0.22)
Income (4000–7499)	0.61*** (0.10)	0.45*** (0.11)	0.82*** (0.10)	0.53** (0.20)	0.85*** (0.14)	0.58** (0.21)	0.48* (0.20)
Income (7500–10000)	0.89*** (0.10)	0.66*** (0.11)	0.96*** (0.11)	1.01*** (0.20)	1.10*** (0.14)	0.73*** (0.22)	0.85*** (0.21)
Income (over 10,000)	0.94*** (0.10)	0.72*** (0.11)	1.01*** (0.11)	1.02*** (0.20)	1.20*** (0.14)	0.87*** (0.22)	0.93*** (0.21)
Observations (N)	5538	5538	5538	2268	5538	2268	2268
R^2/Adjusted R^2	0.13/0.12	0.05/0.05	0.07/0.07	0.07/0.06	0.08/0.07	0.06/0.05	0.07/0.06

Note: Local personal credit scores exist only in cities from C1. Numbers without brackets refer to the linear regression; numbers in brackets are the standard errors of the individual regression coefficients. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.

TABLE A3 | Regression analysis results (2).

Variables	Checking own credit records/scores				Checking others' credit records/scores		
	PBoC reports	SCS lists	Zhima Score	Local credit score (C1)	SCS lists	Zhima Score	Local credit score (C1)
(Intercept)	1.33*** (0.18)	1.27*** (0.18)	2.22*** (0.22)	1.44*** (0.26)	1.23*** (0.16)	1.29*** (0.17)	1.15*** (0.24)
Gender (female)	0.04 (0.03)	0.01 (0.03)	−0.05 (0.03)	0.14** (0.05)	0.01 (0.02)	0.00 (0.03)	0.09* (0.04)
Age (31–40)	−0.03 (0.04)	−0.06 (0.04)	−0.15** (0.05)	0.08 (0.07)	−0.02 (0.03)	−0.10** (0.04)	0.05 (0.06)
Age (41–50)	−0.18*** (0.05)	−0.11* (0.05)	−0.30*** (0.06)	−0.06 (0.08)	−0.08* (0.04)	−0.22*** (0.04)	0.09 (0.07)
Age (51–60)	−0.07 (0.04)	0.02 (0.04)	−0.23*** (0.05)	0.04 (0.06)	−0.01 (0.04)	−0.08 (0.04)	0.18** (0.06)
Age (61–70)	−0.44*** (0.12)	−0.13 (0.12)	−0.70*** (0.15)	−0.17 (0.17)	−0.20 (0.10)	−0.28* (0.11)	−0.09 (0.16)
Age (over 70)	−0.47 (0.31)	−0.36 (0.31)	−0.45 (0.37)	−0.40 (0.48)	−0.50 (0.26)	−0.59* (0.29)	−0.23 (0.44)
Education (high school)	0.07 (0.16)	0.13 (0.16)	0.24 (0.19)	−0.03 (0.21)	0.00 (0.14)	0.02 (0.15)	−0.08 (0.20)
Education (college)	0.12 (0.16)	0.11 (0.16)	0.33 (0.19)	−0.14 (0.21)	−0.05 (0.13)	0.00 (0.15)	−0.03 (0.19)
Education (bachelor)	0.32* (0.15)	0.24 (0.15)	0.39* (0.18)	−0.16 (0.21)	0.08 (0.13)	0.12 (0.14)	0.02 (0.19)
Education (master)	0.45** (0.16)	0.37* (0.16)	0.57** (0.19)	0.08 (0.22)	0.19 (0.14)	0.22 (0.15)	0.13 (0.20)
Education (PhD)	0.36* (0.17)	0.34* (0.17)	0.48* (0.21)	0.14 (0.25)	0.11 (0.15)	0.16 (0.16)	0.07 (0.23)
Income (750–1599)	0.39* (0.15)	0.17 (0.15)	0.21 (0.18)	0.06 (0.23)	0.18 (0.13)	0.42** (0.14)	0.29 (0.21)
Income (1600–2499)	0.45** (0.15)	0.17 (0.15)	0.57** (0.18)	0.13 (0.22)	0.23 (0.13)	0.39** (0.14)	0.18 (0.20)
Income (2500–3999)	0.66*** (0.13)	0.31* (0.13)	0.60*** (0.15)	0.10 (0.19)	0.37*** (0.11)	0.48*** (0.12)	0.23 (0.17)
Income (4000–7499)	0.69*** (0.12)	0.35** (0.12)	0.79*** (0.14)	0.20 (0.17)	0.41*** (0.10)	0.62*** (0.11)	0.33* (0.16)
Income (7500–10000)	0.91*** (0.12)	0.59*** (0.12)	0.91*** (0.14)	0.51** (0.18)	0.62*** (0.10)	0.82*** (0.11)	0.61*** (0.16)
Income (over 10,000)	0.92*** (0.12)	0.60*** (0.12)	0.89*** (0.14)	0.53** (0.18)	0.63*** (0.10)	0.85*** (0.11)	0.58*** (0.16)
Observations (*N*)	5538	5538	5538	2268	5538	5538	2268
R^2/Adjusted R^2	0.05/0.05	0.03/0.03	0.04/0.03	0.05/0.04	0.04/0.04	0.04/0.04	0.05/0.04

Note: Local personal credit scores exist only in cities from C1. Numbers without brackets refer to the linear regression; numbers in brackets are the standard errors of the individual regression coefficients. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.